

IMPROVED DISTRIBUTION SYSTEM PLANNING USING COMPUTATIONAL EVOLUTION

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Abstract—This paper presents a new problem solving environment that can utilize different forms of resources to approach better solutions in the distribution planning domain. To implement this concept, a design optimization framework, named Design By Expectation (DBE), is developed. DBE exploits the adaptation of Genetic Algorithms (GA's) in searching the optimal solution, and releases the strict requirements for objective functions and constraints in conventional optimization techniques. One example, street lighting design, is used to demonstrate the applicability of the proposed approach.

Keywords: Distribution System Planning, Geographic Information System, Design Optimization, Genetic Algorithms, Street Lighting Design

I. BACKGROUND

There are three tasks that every distribution system planning problem needs to solve in terms of topology: *partitioning*, *placement*, and *routing*. Each of them requires an exhaustive search to claim the optimal solution. The exhaustive search is combinatorial with respect to the number of demand/supply sites and feasible route segments, thus making it computationally expensive and infeasible for interactive use.

Five years ago, the authors developed an automated design tool for the Underground Residential Distribution (URD) system, called Automated Electrical Plat Design (AEPD) [1-3]. Based on Intelligent Decision Support System approach, AEPD uses Geographic Information System environment and expert systems to automate the partitioning, placement, and routing tasks in URD planning.

In the early development of the AEPD prototype, the human expertise was implemented in a set of design rules, which helped constrain the search locus, exhibiting the same performance as that of human planners but in a much shorter time frame. To reduce the computational complexity, AEPD divides URD planning into several design modules. The module that determines secondary cable routes, dubbed SER (SEcondary Router) [2], is an example of rule based approach to the distribution system design. The general deficiencies observed in SER are:

- (1) knowledge bases are *specific* based on the company practice,
- (2) knowledge processing becomes a *bottleneck* for SER performance,
- (3) resulted solutions *may not be better* than manual designs, and
- (4) SER *does not seek the local optimality* for cable routing and device placement.

The AEPD module for the optimal routing of primary distribution systems is called the Automated Primary Router (APR) [4], which employs a heuristic search algorithm. Although the APR provides a local optimal result for the given device placement, the global optimality can not be guaranteed because of the missing interaction with device placement modules. In AEPD, the collaboration between the cable routing and device placement modules are achieved only through the user interaction, which prohibits the full automation and global optimization. In addition, having the user as the link between the automated placement and routing modules restrains the quality of the design to the user's knowledge and expertise.

The new approach to distribution planning proposed in this paper is driven by the goal of improving the performance of planning operators (design tasks) with human knowledge, spatial heuristics, and computational evolution. The optimization of complex distribution planning normally constructs a problem space where several local maxima co-exist with the global maximum (multi-modal functions). To find the extrema of such a large space is often a NP-Hard problem.

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Traditionally, a rule-based system can reach, without tackling with the combinatorial explosion, the same point where a human expert can; however, it has no provision to evolve a better solution beyond the knowledge that it contains. Hill climbing, based on the extracted heuristics or a set of algorithms, tends to efficiently reach an optimal point without knowing it is global or local (for a complex problem). With informative heuristics and carefully-defined cost function, hill-climbing can be very efficient and accurate. Newton's method (e.g., optimal power flow problems) and heuristic search algorithms all fall into this category. However, the informative heuristics are difficult to find and they lead to local optima, more often than not, when the starting point is far from the extrema.

An adaptive search with implicit parallelism, on the other hand, is able to make use of the knowledge found at points previously evaluated, and explore unknown space points in an intelligent way. The former capability is referred to as *exploitation*, while the latter is called *exploration*. Genetic algorithms (GA's) is such an example. Holland [5] showed that a GA combines both *exploration* and *exploitation* simultaneously in the optimal way in spite of several assumptions.

II. GENETIC ALGORITHMS

Genetic Algorithms (GA's), inspired by natural evolution, have been successfully applied to a set of numerical optimization problems [7-10]; with the superior parallelism and stochasticity, GA's are suitable for solving complicated optimization problems, such as those encountered in electric distribution planning [11].

GA's can be described as iterative procedures that maintain a population P of n candidates to an objective function f :

$$P(t) = \{x_1(t), x_2(t), \dots, x_n(t)\} \quad (1)$$

Each structure $x_i(t)$ in population $P(t)$ at time t is simply a binary string of length l . Generally, each x_i represents a vector of parameters to the function $f(x)$, but the meaning associated with the vector is unknown to the GA. During each iteration (one generation), the current population is evaluated, and, on the basis of that *evaluation*, a new population of candidate solutions is formed.

The initial population $P(0)$ is usually generated at random. Alternatively, the initial population may contain heuristically chosen initial points—*Seeds*. In either case, the initial population should contain a wide variety of structures, revealing enough information about the search space. Each structure in $P(0)$ is then evaluated.

The structures of the population $P(t+1)$ are chosen from the population $P(t)$ by a randomized selection procedure that ensures that the expected number of times a structure is chosen is proportional to that structure's performance,

relative to the rest of the population. At the end of the selection procedure, the population $P(t+1)$ contains the exact duplicates of the selected structures in population $P(t)$. The whole process is called *reproduction*.

To search other points in the search space, some variation is introduced into the new population by means of idealized genetic operators: *crossover* and *mutation*. Under the crossover operator, two structures in the new population exchange portions of their genes (in a form of binary representation). After a *crossover* operation, a *mutation* operator may apply to randomly explore other patterns. Figure 1 shows the operation of *crossover* and *mutation*. After the process of *reproduction*, *crossover* and *mutation* is completed, the next population is evaluated. The process of *evaluation*, *reproduction*, *crossover*, and *mutation* forms one generation in the execution of a GA.

Because GA's transform the problem space to a generic coding scheme, they do not need any special forms of objective functions and constraints. The objective functions and constraints could be linear/nonlinear, or continuous/discrete. Based on the generic coding scheme, GA's form a new search space. Holland [5] developed a *schemata* theorem to explain how the *crossover* and *mutation* operations effectively sample the new search space. The efficiency is of the order n^3 , where n is the population size. This property is known as *implicit parallelism* [5], which proves the good performance of GA's.

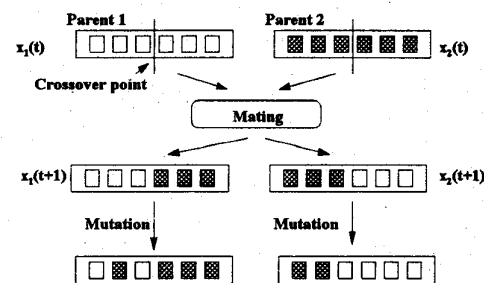


Figure 1 Crossover and Mutation Operations

III. DESIGN BY EXPECTATION

To implement the concept of the proposed approach, a framework called *Design By Expectation* (DBE) [6] is utilized in this paper.

DBE exploits the adaptation of GA's in approaching better solutions, which compensates for the weak creativity of traditional computation. By developing a set of expectation interpreters that release the strict requirements on the formalities of objective functions and constraints required in traditional optimization techniques, the DBE framework becomes more flexible.

The DBE framework is a goal-oriented implementation, which takes advantage of adaptation in GA's and domain-

pre-determined, the investment cost mainly depends on how the routes are determined, which, in turn, is based on where the streetlights are placed. The total complexity of the streetlight placement is of the order given by:

$$O(n, m, p, q, s) = \frac{n!}{m!(n-m)!} \times p^q \times s \quad (2)$$

where n = number of feasible supply sites,
 m = pre-determined number of lights,
 p = average number of arcs incident to a node,
 q = average number of segments making up a route, and
 s = number of possible supply sites.

Equation (2) clearly demonstrates the combinatorial nature of street lighting design. Because the objective is difficult to formulate in a mathematical form, this problem is ideally suited for applying the DBE framework to evolve a better solution.

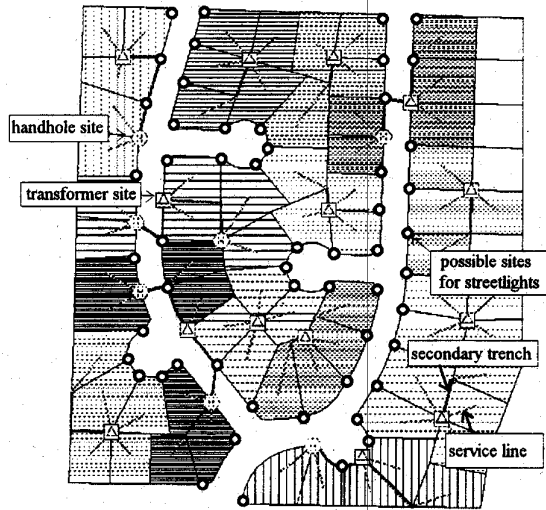


Figure 4 Completed Plat Design without Street Lights

B. System Structure

The collaboration between GA's and an intelligent router (similar to the one used in primary distribution systems [4]) is the core of the street lighting design. Figure 6 illustrates the system structure of such a scheme.

The GA's first randomly place the streetlights along streets or incorporate the design results from a human designer. Then, a parallel search starts with the assistance of the intelligent router, approaching the point where the performance is the best. The intelligent router can provide solutions for the shortest cable routes once the streetlights are placed; this gives a good measure of the minimal investment expectation.

Different expectations are separately interpreted. A set of weighting factors is determined according to the cost

requirements and company practices. The sum of the weighted and quantified expectation is transformed into a performance index (PI), which is then used to measure the quality of the final design solution.

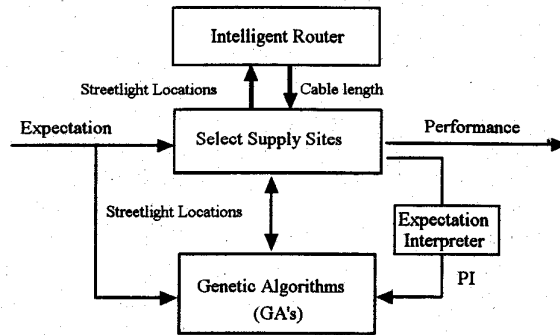


Figure 5 DBE Based Street Lighting Design

C. Expectation Interpreters

The minimal investment cost is easy to quantify with the help of the intelligent router [4]. However, the other objectives are difficult to quantify. In this paper, several expectation interpreters are developed utilizing the spatial relationship among different objects to reduce the computational burden. The quantitative interpretation of the expectation for *evenly placed streetlights* is described below. Consider a street layout shown in Figure 6 (a). Assume one streetlight (with a specific luminaire) can illuminate all its adjacent customers (lots) and those customers adjacent to its connected nodes. The check marker in Figure 6 (a) indicates all the customers who have adequate light level after a streetlight is placed. If the total number of customers along the streets is c , and total number of customers who have adequate lighting is $c' = f(st_i)$ (st_i : the i th street light, $i = 1, 2, \dots, m$), then an index for the quality of the street lighting design can be defined as:

$$Q(c, c')_{\text{street-lighting}} = k \times \left(\frac{c'}{c} \right) \quad (3)$$

where k is a weighting factor.

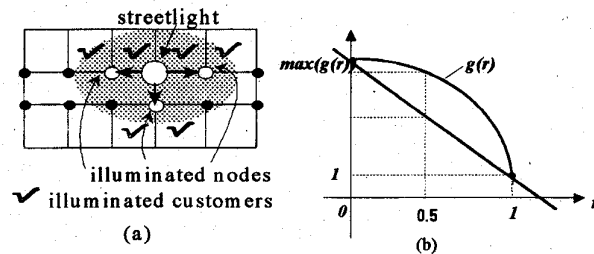


Figure 6 (a) Luminous Effect on a Street Layout
 (b) Nonlinear Expectation $g(r)$

If the ratio $r = c'/c$ is close to 1, the streetlights are evenly placed along the streets (given the customers (lots) are evenly distributed on the development). The weighting factor k can be a constant or a function of c and c' . In many cases, the expectation to interpret the r is in a nonlinear form. For example, $r = 1$ is twice more desirable than $r = 0.5$. Hence, $k = k_1 \times g(r)$ is defined to reflect the nonlinear expectation; $g(r)$ in Figure 6 (b) is chosen to magnify the resolution between a good solution ($r = 1$) and others.

The $g(r)$ is expected to give a good resolution between an evenly placed street lighting design and an erratically placed street lighting design. k_1 is to provide a flexibility to collaborate with other expectations. The expectation for other possibilities, such as all the streetlights on the same street side, can be interpreted in the similar manner. Assuming the total number of street sides in the development is d , the street lights are placed on d' sides of streets, a ratio r' can be found as:

$$r' = \frac{d'}{d}, \text{ where } d' \in [1, d] \quad (4)$$

To expect all the street lights on the same side of streets, r' should be close to $(1/d)$. A similar expectation index can be defined to distinguish the case of placing all streetlights on the same street side from others. Again, a weighting factor should be applied to provide a flexibility for further collaboration with other expectation interpreters.

D. Implementation

A binary encoding scheme is used to form a chromosome string in the GA's. A chromosome string contains the combination encoding of different decision variables (known as *genes*). In the street lighting design, each feasible supply site (for streetlights) is assigned as a gene with only one bit as its value. The value 1 indicates the feasible site is placed with a streetlight, while the value 0 means no streetlight is placed. Figure 7 shows such an arrangement.

The development of the GA modules is based on the GENITOR package [8]. The genetic operators are modified to guarantee the new solution sets are always feasible in this specific example. For instance, the mutation operator only performs (0 1) pair mutation, meaning that mutating a "0" to "1" incurs a counter-mutation "1" to "0" at the other bit. The production of first generation is designed to deliver only the feasible solution sets in the population pool. With the feasible first generation and specially designed operators, the implemented GA procedures ensure each generation contains only feasible solution sets, avoiding the massive disturbances caused by infeasible solution sets along the evolution process.

The implementation of the expectation interpreter for even placement is based on equation (3). Other expectation interpreters can be implemented in the same way, depending

on the objective and the distribution engineer's preferences, such as equation (4). The intelligent router, GA's, and expectation interpreter modules are all implemented in C programming language. GA's first produce a set of random locations for streetlights and send the information to the intelligent router. The intelligent router will connect each streetlight to the closest supply site. Then, the different expectation interpreters work together to evaluate each solution set according to the routing and placement results. Finally, each solution set will have a performance index (the evaluation result) as the fitness measure in the evolution process.

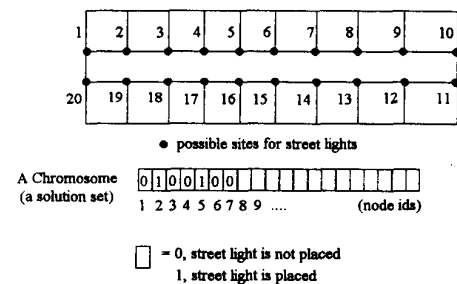


Figure 7 Encoding Scheme for Street Lighting Design

E. Performance Result

Table 1 shows the result of three different cases, in all of which 20 streetlights are to be placed on the specified residential development. The result demonstrates the feasibility of the DBE-based street lighting design. The street lighting design is currently carried out manually by customer engineers in electric utilities, which normally takes several hours to complete the first workorder. Even more time will be needed to improve the design further. The efficiency of the DBE based street lighting design meets the requirements as an *interactive* design tool, which enables a *thorough* evaluation of different design practices.

Table 1 Performance Result (based on 5000 generations, population size = 350, and mutation rate = 0.3)

Case No.	Objectives	Cost (\$)	Time (sec.)
1	(1)	5430.95	12.1
2	(2)	5789.95	11.2
3	(1)+(2)	5519.45	19.2

(Note: (1) = minimal cable length, (2) = even placement)

Figure 8 shows the result with the minimal cable length (case 1). It is obvious that all the streetlights are placed either at/near a transformer site or a handhole site to minimize the cable length. If the streetlights are evenly placed, the total cost is higher because the length of cables is not optimized. Figure 9 is such an example (case 2). Figure 10 shows a design that resolves the multi-objectives (case 3):

minimal cable length and even placement, using the appropriate weighting factors. Table 1 only shows the cost comparison for these three cases because the interpretation of other objectives is subjective, depending on how the expectation interpreters are implemented. The intent of Table 1 and the corresponding figures is to show how DBE framework can easily resolve the multi-criteria optimization without imposing a significant increase in the execution time.

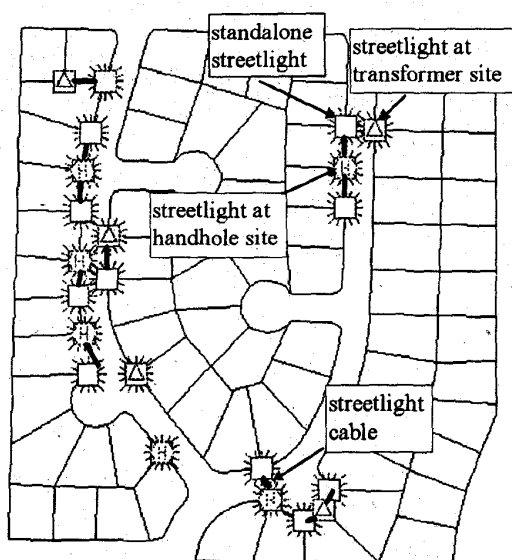


Figure 8 Case 1: Streetlight Design with Minimal Cable Length

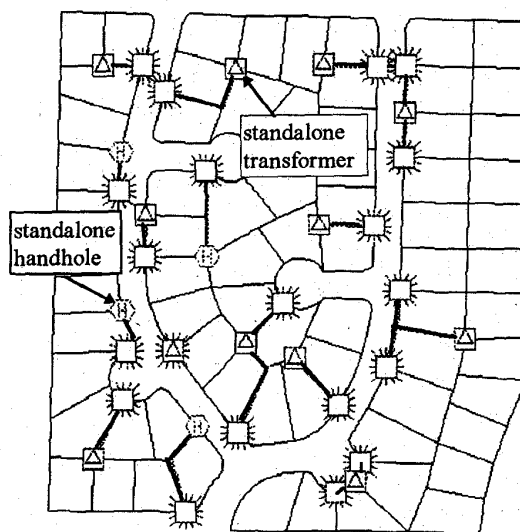


Figure 9 Case 2: Streetlight Design with Even Placement of Lights

V. CONCLUSIONS

In this paper, the concept of the *optimal distribution planning using computational evolution*, and the *Design By Expectation (DBE)* framework that improves the distribution subsystem design have been presented. The collaborative scheme of the DBE framework hybridizes the domain-specific knowledge (expectation interpreters), adaptation of GA's, and other optimization techniques (heuristic search) to enable the interaction between the routing and placement modules. With the example of street lighting design, the DBE framework demonstrates an automated way of improving the quality of distribution planning, which paves a way for eventual global optimization in the distribution system planning domain.

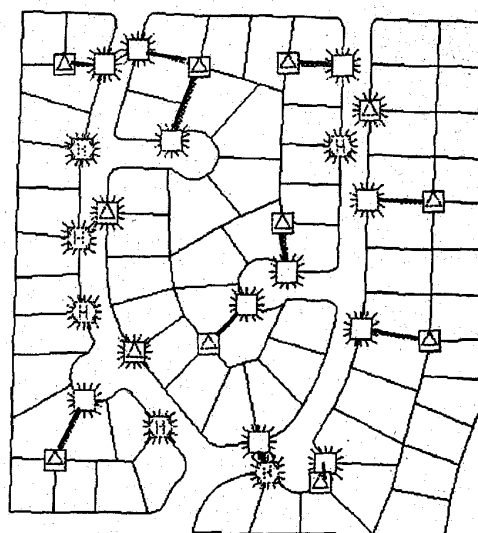


Figure 10 Case 3: Streetlight Design with Minimal Cable Length and Even Placement of Lights

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VII. BIOGRAPHIES

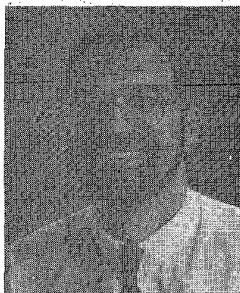


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